CSCI 3161 – Computer Animation

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About the course…

What this course is not

- A tutorial on using computer animation software
- An art class
- A bleeding edge course for experienced graphics developers

What this course is

- An **introduction** to **established** animation software methods
- A practical introduction to OpenGL programming

About the course…

- Computer animation deals with all aspects of creating moving images with a computer:

  - –Programming
  - –Modeling
  - –Animating
  - –Lighting
  - –Rendering
  - –Post-processing
  - –Scripting
  - –Storyboarding
  - –Layout
  - –Hardware
  - –Software
  - –Sound

About the course…

- However, we will only be concerned with certain technical aspects of:
  - –Programming
  - –Modeling
  - –Animating
  - –Lighting
  - –Rendering
  - –Post-processing
  - –Scripting
  - –Storyboarding
  - –Layout
  - –Hardware (Shaders)
  - –Software
  - –Sound

About the course…

Pre-requisites

- Some experience with C programming
- CSCI 2110 (Computer Science III)
- CSCI 2132 or CSCI 2134 (Software Development)
- MATH 1030 (Matrix Theory and Linear Algebra 1)

What you don't need to know

- OpenGL programming
- Computer graphics
- Art, drawing, etc...

About the course…

Expectations

- This is an introductory course (not a Crytek tech talk)
- **Not** designed for someone already proficient in graphics programming
- We have to start with traditional OpenGL 1.4
- But later in the course get into **some** shaders and GPU stuff
- Take 4160 for more advanced graphics algorithms

About the course…

Why not start with Shaders and the latest/greatest stuff?

- Again, this is an introductory course
- Shaders and GPU programming adds a layer of complexity for learners
- Too much plumbing details to learn before you can do anything
- The material is challenging enough as it is
- We will work up to shaders later in the course

About the course…

Why not use OOP?

- Its possible to use C++ but, unless you are careful you will:
  - bloat your memory use
  - write slow code
- For graphics, speed is everything:
  - Don't want to create zillions of expensive code objects
  - Requires more than just basic C++ knowledge to avoid these pitfalls
- Its more straightforward to write faster code in C

About the course…

Hardware for Assignments

- *Ideally*, use a machine with decent graphics
  - though, *doesn't have to be top of the line*
- *E*ven an onboard graphics chip with shared memory on a laptop might be ok these days...
- Probably not the lowest end laptop you can find (Chromebook, etc)

About the course…

Software for Assignments

- Windows platform
- MS Visual Studio
- Use the OpenGL and FreeGLUT libraries
- See www.cs.dal.ca/~sbrooks/csci3161/reference/index.html
  - for example Visual Studio Projects
  - and other help getting started

About the course…

Course Webpage

Description

The course provides a comprehensive overview of algorithms and software techniques for generating animated movies and interactive animation programs. This course is aimed at computer science students interested in understanding the underlying concepts and techniques of computer animation. Through the development of a significant project using industry standard graphics libraries, students will learn proven techniques of computer animation. Though not extensively mathematical, the course will expose students to mathematical concepts such as curve representations, numeric interpolation and parameterization where appropriate.

Login: dal **Password:** user

About the course…

Working on your own machine

- Can use C, FreeGLUT and OpenGL on your own machine
- But, other platforms will not be directly supported
- See: www.cs.dal.ca/~sbrooks/csci3161/reference/index.html

About the course…

Class lectures:

- Wednesdays & Fridays
- 10:05 - 11:25
- Carleton TUPPER BLDG THTR C

Getting Help in the Learning Center

- Where:
  - **In-person** in the learning center
- When:
  - Tuesdays, 3pm - 4pm
  - Thursdays, 1pm - 2pm
- What:
  - Meeting place for students
  - Get help from our TA (Ben Lee)
  - No formal instruction

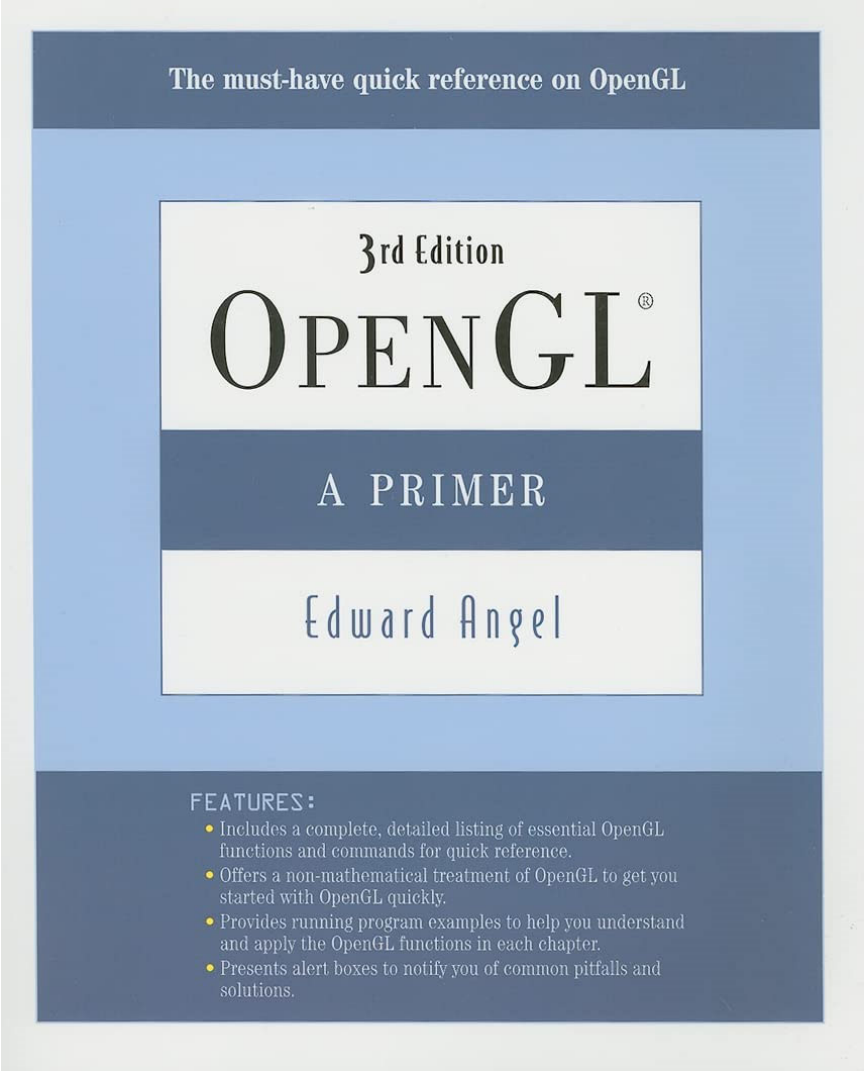
Office Hours

- **Online** TA Office Hours:
  - Get even more help from our TA, One-on-one
  - When:
    - Wednesday, 3pm - 4pm
  - Where:
    - Online on Brightspace
- My **in-person** office Hours:
  - For higher level course related issues
  - When:
    - Friday, 2pm-3pm
  - Where:
    - Mona Campbell Rm 4206

About the course…

Optional Course Text

- **OpenGL: A Primer**
- Kindle or Paperback
- Can get cheaper used on Amazon!



Other Optional Reference Texts

Math review:

- **Computer Graphics: Mathematical First Steps**, William Hall, Patricia Egerton and Pat Egerton, Prentice Hall.

Alternate Textbooks:

- Computer Animation: Algorithms and Techniques, Rick Parent, Morgan-Kaufmann.
- Computer Graphics Using Open GL, Francis S. Hill. Prentice Hall.

Advanced & Interesting Books:

- Advanced Animation and Rendering Techniques: Theory and Practice, Alan Watt and Mark Watt. Addison-Wesley.
- Texturing and Modeling: A Procedural Approach, David Ebert, F. Kenton Musgrave, Darwyn Peachey, Ken Perlin and Steven Worley. Morgan Kaufmann.

About the course…

Grading

- Tests (40%)
  - Two **in-class** tests, each worth 20%
- Programming Assignments (20%)
  - Two assignments, each worth 10%
- Final Project (40%)

Late Submission Policy

- Late work will be penalized 5% per day (or part thereof)
- You may not receive credit for work that is more than 3 days late
- The Final Project CANNOT be handed in late

Test Dates

Test 1 (in class):

- Oct 9th

Test 2 (in class):

- Nov 27th

About the course…

Academic Honesty

- All work you submit must be your own
- OK to discuss problems
- Materials you hand in must be done individually
- Any materials referenced must be attributed
- No use of AI is permitted for generating code
- *Assignments and projects will be checked with automated software*

Never:

- show another student code that you have written for an assignment
- or, write code for another student
- or, copy code from/to another student

Course Topics

1. Introduction to Computer Animation (CA)
   - History of CA
   - Applications of CA
   - Methods of CA
   - The rendering pipeline

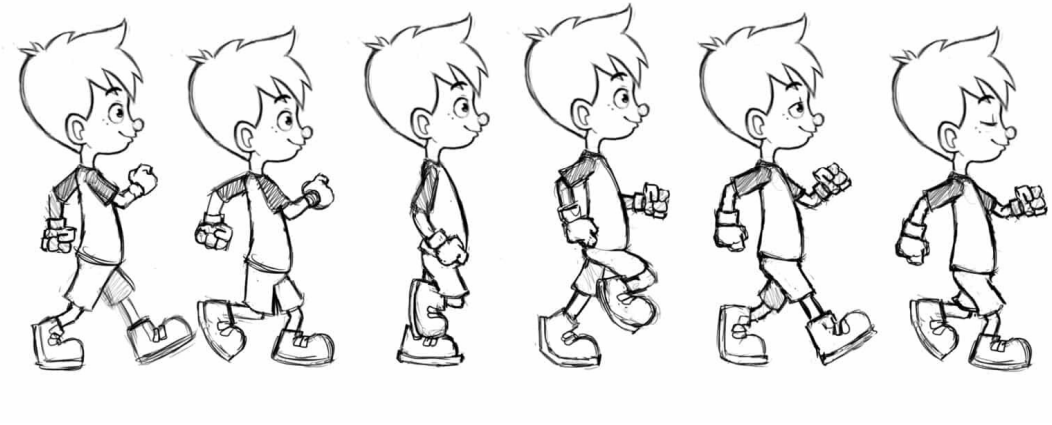
Course Topics

2. Introduction to OpenGL Programming
   - The OpenGL architecture
   - Functions
   - Compiling a simple program

3. Basic I/O for Graphics
   - The event queue
   - Callbacks
   - Menus
   - Mouse/keyboard input

Course Topics

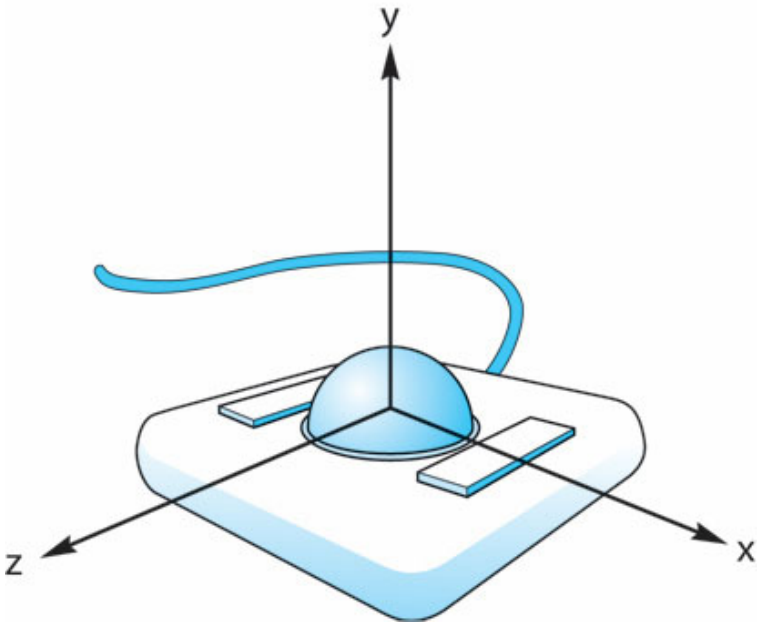
4. 2D Animation Basics
   - Geometry review
   - Double buffering
   - Interactive programs



5. Interpolation
   - Linear interpolation
   - Temporal interpolation

Course Topics

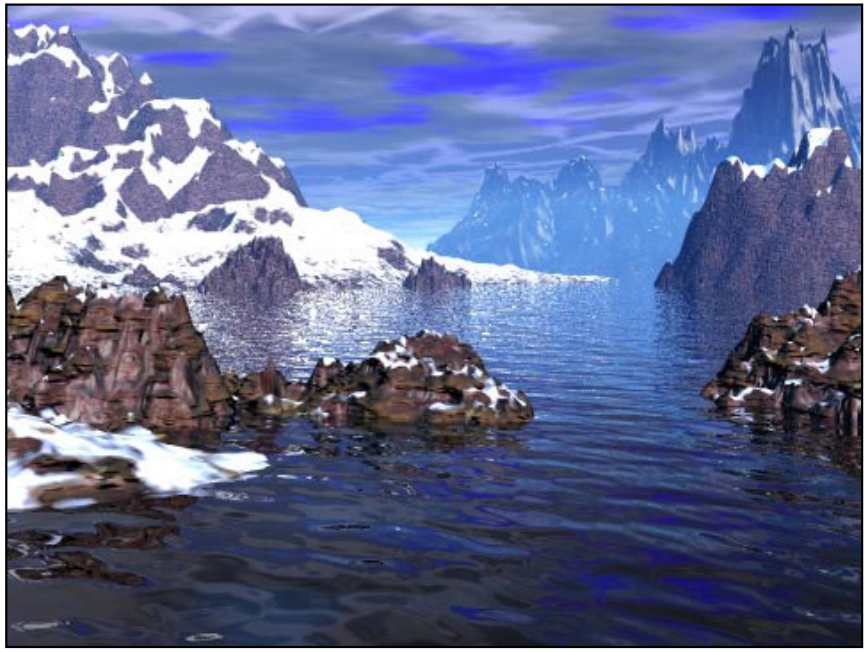
6. 3D Animation Basics
   - Coordinates and cameras
   - OpenGL Transformations
   - OpenGL projections



Course Topics

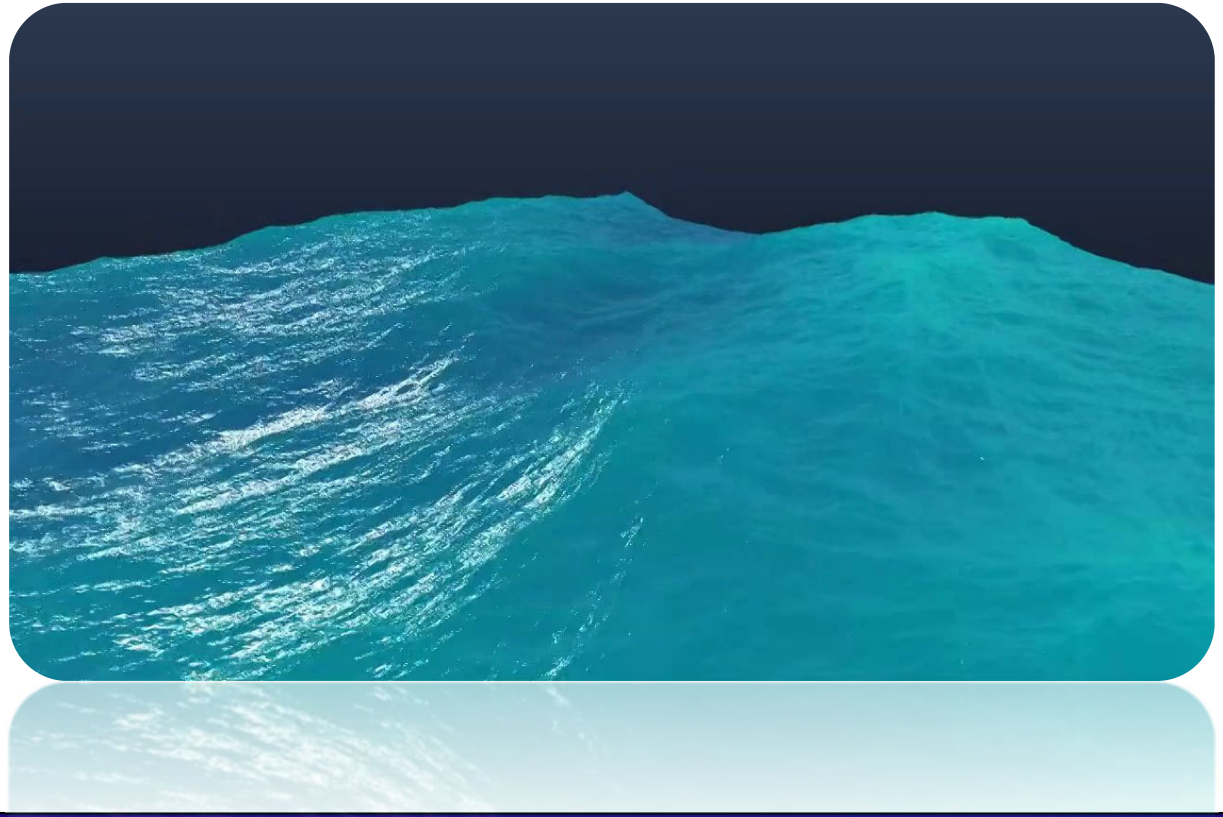
7. Surfaces and Lighting
   - Polygonal models
   - Shading in OpenGL
   - Texture mapping

8. Procedural Graphics (maybe)
   - Procedural texturing
   - Animated surfaces
   - Flocking behavior



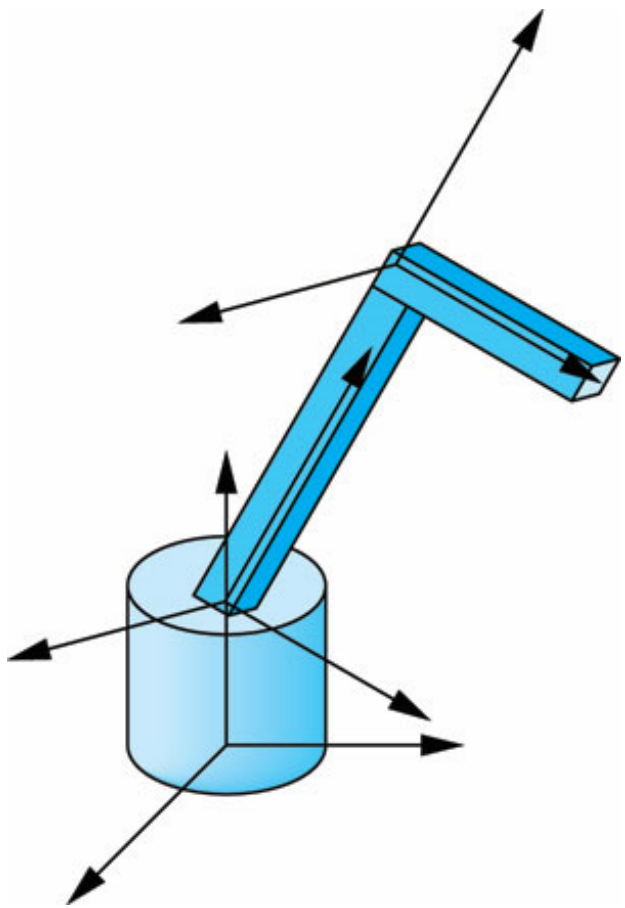
Course Topics

9. OpenGL GLSL
   - Introduction to shaders
   - Vertex shaders
   - Pixel shaders



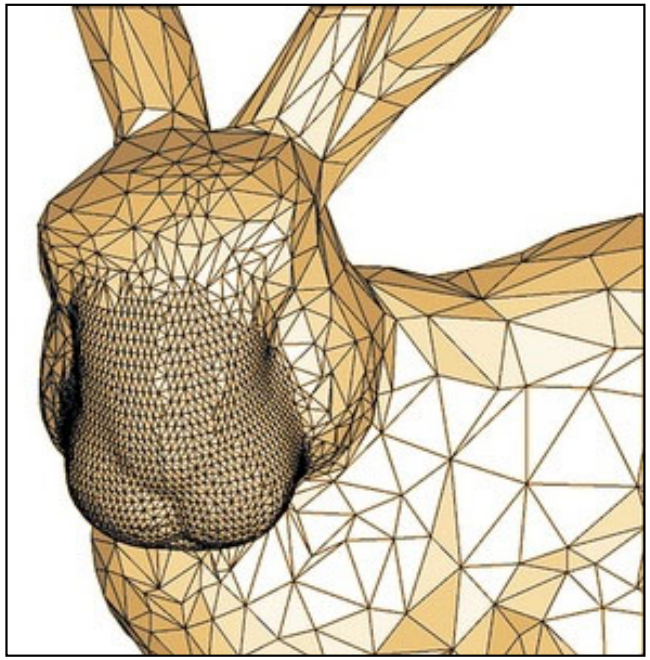
Possible Course Topics

10. Hierarchical Animation
    - Articulated models
    - DAG models
    - L-Systems



Possible Course Topics

11. Pragmatic Issues
    - Level-of-detail



12. Post-processing
    - Compositing
    - Color processing
    - Anti-aliasing
    - Motion blur

Guest Talks

Home Work

- Check out the reference page:
  - http://web.cs.dal.ca/~sbrooks/csci3161/reference/index.html
- See if you can get one of the simple test programs to compile on your home system

Title Slide Images from Canadian Researchers

Image references, clockwise from top left:

- liu, Hsueh-Ti Derek, and Alec Jacobson. “Cubic Stylization.” ACM Transactions on Graphics 38.6 (2019): 1–10.
- Luc Leblanc, Jocelyn Houle, and Pierre Poulin. 2011. Component-based modeling of complete buildings. Graphics Interface. Canadian Human-Computer Communications Society, Waterloo, CAN, 87–94.
- Graham LeBlanc, Andrew Shouldice, Dirk V. Arnold and Stephen Brooks. Multi-band Fourier synthesis of ocean waves. Journal of Graphics Tools, volume 16, issue 2, pp. 57-70, 2012.
- Adrian Blumer, Jan Novák, Ralf Habel, Derek Nowrouzezahrai, Wojciech Jarosz. Reduced aggregate scattering operators for path tracing. Computer Graphics Forum, 35(7):461–473, October 2016.
- Fang Da, Christopher Batty, Chris Wojtan, and Eitan Grinspun. 2015. Double bubbles sans toil and trouble: discrete circulation-preserving vortex sheets for soap films and foams. ACM Trans. Graph. 34, 4, 9 pages.

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The Making of …